Estuaries and Coasts (2008) 31:1208–1220
DOI 10.1007/s12237-008-9098-8

# Seston Removal by Natural and Constructed Intertidal Eastern Oyster (*Crassostrea virginica*) Reefs: A Comparison with Previous Laboratory Studies, and the Value of in situ Methods

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Received: 15 November 2007 / Revised: 9 July 2008 / Accepted: 29 August 2008 / Published online: 23 September 2008
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**Abstract** An important ecological role ascribed to oysters is the transfer of materials from the water column to the benthos as they feed on suspended particles (seston). This ecosystem service has been often touted as a major reason for many oyster restoration efforts, but empirical characterization and quantification of seston removal rates in the field have been lacking. Changes in chlorophyll *a* (chl *a*) concentrations in the water column were measured in May 2005 and June 2006 in South Carolina using in situ fluorometry and laboratory analysis of pumped water samples taken upstream and downstream as water flowed over natural and constructed intertidal oyster reefs. Both methods gave similar results overall, but with wide variability within individual reef datasets. In situ fluorometer data logged at 10 to 30-s intervals for up to 1.3 h over eight different reefs (three natural and five constructed) showed total removal (or uptake) expressed as % removal of chl *a* ranging from −9.8% to 27.9%, with a mean of 12.9%. Our data indicate that restored shellfish reefs should provide water-quality improvements soon after construction, and the overall impact is probably determined by the size and density of the resident filter feeder populations relative to water flow characteristics over the reef. The measured population-level chl *a* removal was converted to mean individual clearance rates to allow comparison with previous laboratory studies. Although direct comparisons could not be made due to the small size of oysters on the study reefs (mean shell height, 36.1 mm), our calculated rates (mean, 1.21 L $h^{-1}$) were similar to published laboratory measured rates for oysters of this size. However, the wide variability in measured removal by the oyster reefs suggests that individual oyster feeding rates in nature may be much more variable than in the laboratory. The proliferation of ecosystem-level models that simulate the impacts of bivalves on water quality based only on laboratory-feeding measurements underscores the importance of further research aimed at determining ecologically realistic feeding rates for oysters in the field. Because in situ methods provide many replicate measurements quickly, they represent a potentially powerful tool for quantifying the effects of oyster reefs, including all suspension-feeding taxa present, on water quality.

**Keywords** Oyster reef · Bivalve · Water quality · Fluorometry · Ecosystem services

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## Introduction

One of the major ecological functions of suspension-feeding bivalve molluscs such as oysters is their potential to improve water quality by filtering plankton and other

suspended particles (seston) from the water column (Dame 1996; Kennedy 1996; Harsh and Luckenbach 1999, Dame et al. 2001; French McCay 2003; Newell and Koch 2004). Ongoing and planned oyster restoration projects in many areas emphasize ecosystem services such as water-quality improvements, instead of or at least in addition to, the historical role of oysters as a commercial resource (Luckenbach et al. 1999, 2005; Brumbaugh et al. 2000; Coen and Luckenbach 2000, Coen et al. 2007; Grabowski and Peterson 2007). Quantification of seston removal rates, however, has, until recently, been based mainly on laboratory studies. Previous field studies typically have not demonstrated measurable seston uptake by natural (e.g., Dame and Libes 1993; Wilson-Ormond et al. 1997) or restored (Nelson et al. 2004) oyster reefs. Cressman et al. (2003), however, measured up to 25% decreases in chlorophyll *a* (chl *a*) at times downstream of small intertidal reefs in North Carolina. A comparative field study involving three bivalve species found maximum measured seston removal using field sampling with in situ fluorometers as water flowed over the shellfish bed or reef ranging from 27.8% for mussels (*Mytilus* spp), 37.4% for eastern oysters (*Crassostrea virginica*), to 62.3% for hard clams (*Mercenaria mercenaria*), with the differences in removal mainly attributed to differences in bivalve size and densities relative to water flow and water depth (Grizzle et al. 2006). These recent studies, therefore, indicate oysters should clear substantial quantities of water, as has been demonstrated by lab (e.g., Harsh and Luckenbach 1999) and field studies on several other bivalve taxa (Judge et al. 1993, Alpine and Cloern 1992; Haamer 1996; Coen et al. 2000; Haamer and Rodhe 2000; Ackerman et al. 2001; Grant et al. 2007). The recent studies also corroborate earlier research that found substantial seston removal at times as water flowed through a submerged field flume (tunnel) over intertidal oyster reefs (see Dame et al. 1992, Dame 1996).

Ecosystem-level models or other assessments that involve eastern oyster (*C. virginica*) feeding rates (e.g., Newell 1988; Newell et al. 2005, 2007; Pomeroy et al. 2006; Cerco and Noel 2007) have been based on the few laboratory measurements available (primarily Jordan 1987). There is a need to expand this research to include field measurements of feeding rates over a wide range of environmental conditions (Powell et al. 1992; Harsh and Luckenbach 1999). It is well established that various environmental factors affect the feeding behavior of suspension-feeding bivalves in general (see reviews by Newell and Langdon 1996; Shumway 1996; Grizzle et al. 2001; Peterson 2001) and the natural environment is typically much more variable than is practical to simulate in the laboratory (Judge et al. 1993). Therefore, it seems reasonable to expect that field studies may yield quite different feeding rates compared to laboratory studies.

In the present study, we characterize short-term (up to 1.3 h) seston removal by natural and constructed ("restored") intertidal eastern oyster reefs in South Carolina measured by in situ fluorometry and laboratory analyses of pumped water samples using extraction and fluorometry. We also compare our field data to previous laboratory studies that have been used to infer the impacts of oyster filtration on various ecosystem-level processes, and discuss the value of in situ methods for future research.

## Materials and Methods

### Study Sites

All sampled oyster reefs were intertidal in estuaries along coastal South Carolina, USA. Measurements were made over five different constructed reefs during 16–19 May 2005, and three natural reefs within the ACE Basin National Estuarine Research Reserve (ACE NERR) on 7–8 June 2006 (see Table 1 for summary information). Most of these reefs were in areas with a 'Restricted Harvesting' status based on SC-DHEC's management criteria.

### Field and Laboratory Methods

The general protocol for field studies consisted of making repeated measurements of environmental conditions and chl *a* concentrations upstream and downstream on each reef using in situ fluorometry and laboratory analysis of pumped

**Table 1** Location of study reefs

| Sampling Date | Reef Name | General Location | Latitude/Longitude |
|---|---|---|---|
| 5/16/2005 | Woodland Ck (2000) | Murrells Inlet | 33.53933N 79.03350W |
| 5/17/2005 | Oak Ck SCORE (2003) | Murrells Inlet | 33.51833N 79.05833W |
| 5/17/2005 | Oak Ck SCORE (2001) | Murrells Inlet | 33.51833N 79.05833W |
| 5/19/2005 | Clam/Oyster (2000) | ACE Basin, Twin Sisters Ck | 32.51263N 80.43878W |
| 5/19/2005 | Recycled Shell (2000) | ACE Basin, Twin Sisters Ck | 32.51444N 80.43766W |
| 6/7/2006 | Store Ck#1-Natural | Store Ck | 32.55286N 80.33284W |
| 6/7/2006 | Store Ck#2-Natural | Store Ck | 32.55210N 80.33246W |
| 6/8/2006 | Bailey Ck-Natural | Bailey Ck, Bailey Island | 32.55431N 80.36105W |

Year in parentheses indicates time of construction for restored reefs

water samples (previously described in detail by Grizzle et al. 2006). Each of the fluorometer probes and water intake tubes was set 5 to 10 cm above the top of the reef. Fluorometry data were continuously logged at 10 to 30-s intervals for the duration of each measurement period. Readings from the two in situ fluorometers in most cases were compared side-by-side at the beginning of each deployment and again after the last set of readings was taken to be sure they gave similar readings. Two to five pumped water samples were also taken during each fluorometer sampling period for laboratory analysis of chl *a* concentrations and comparison to the in situ fluorometry data.

Pumped water samples were stored in the dark on ice and returned to the laboratory for filtration (Gelman GF/C filters) within 6 h of collection. The filters were then analyzed fluorometrically after 90% acetone extraction following a modified EPA Method 445.0 procedure using a calibrated Turner 10-AU and the acidification method (Arar 1997, Rev. 1.2).

Oyster size and density on each reef was determined based on replicate ($n$=4 to 6) 0.07 or 0.1 $m^2$ quadrat samples. In some cases, data were available from recent sampling at a site using larger replicate sample and quadrat sizes. All bivalves (primarily oysters, with some mussels [*Geukensia demissa*, *Ischadium recurvum*, *Brachidontes exustus*] and a few clams [*M. mercenaria*]) were counted and measured (shell length or height where appropriate to the nearest mm using dial calipers).

Flow length, which was the distance between the two in situ fluorometers, was measured to the nearest 0.5 m. Water depth was measured to the nearest cm at 5 to10-min intervals using a marked stadia rod. Water flow speed was measured at the same height (5 to 10 cm above the bottom) as the fluorometer probes with a Marsh–McBirney Model 201 electromagnetic current meter at 5 to 10-min intervals for the duration of each measurement period. Water flow direction was determined several times during each measurement period by releasing a small floating object upstream of the reef and fluorometer to be sure that the fluorometers were correctly positioned. Salinity and temperature data were obtained either directly from the pumped water samples with a refractometer or from the ACE NERR SWMP (System Wide Monitoring Program) water monitoring site in St. Pierre Creek near the five study reefs in the ACE Basin; no extended water-quality data were available from the three sites nearest Murrells Inlet, Georgetown County in 2005.

## Calculations

Seston (chl *a*) removal by the bivalve populations (mainly oysters) on each study reef was expressed as % of the chl *a* removed from the water column using the following where $C_1$=upstream chl *a* concentration, and $C_2$=downstream chl *a* concentration.

$$\%\ \text{Chl } a \text{ removal} = (C_1 - C_2)/C_1 \times 100 \quad (1)$$

This equation was used to calculate overall removal for each individual reef dataset, and at 5-min intervals using the in situ fluorometry data to quantify short-term variations in removal that could be directly compared to the pumped water samples (which required ~5 min/sample) for laboratory chl *a* analysis. This equation is essentially the "traditional" method developed by Coughlan (1969) and typically used to measure filtration rates of bivalves in laboratory flow-through chambers where seston concentrations in the outflow from a control chamber with no bivalve are compared to seston concentrations in the outflow from the chamber with bivalve (Navarro and Velasco 2003). Equation 1 expresses the difference between the two measurements (our upstream vs. downstream) as a percentage of chl *a* removed.

The population-level bivalve data from Equation 1 were converted to average individual clearance rate ($CR_{ind}$; L $h^{-1}$) using the following equation where $A_{xsec}$ = cross-sectional area of water column ($m^2$); $v$ = water flow speed (m $h^{-1}$); $N$ = mean bivalve density (# of individuals $m^{-2}$); $A_{reef}$ = reef surface area ($m^2$); and % chl *a* removal is from Equation 1.

$$\text{CR}_{\text{ind}} = ([A_{\text{xsec}} \times v \times 1,000]/[N \times A_{\text{reef}}]) \times (\%\ \text{chl}\, a\, \text{removal}/100) \quad (2)$$

This equation essentially expresses the measured seston uptake by the reef as a whole (as determined using Eq. 1) in terms of the average individual bivalve clearance rate.

The following two equations were used to allow a comparison of our field-derived calculated individual feeding rates (Eq. 2) to individual feeding rates from previous laboratory studies, which were expressed in soft tissue dry weight units instead of shell height that we measured directly (but see Harsh and Luckenbach 1999). Equation 3 is a power function developed for South Carolina intertidal oysters based on a sample of ~300 individuals from three different reefs sampled during 2007 that allowed conversion of our shell height measurements to soft tissue dry weight (DW; g).

$$\text{DW} = 0.00003 \times \text{shell height}^{2.1727} \quad (3)$$

Equation 4 was developed by Riisgård (1988) for a wide range of oyster sizes feeding in the laboratory under near optimal temperatures (27 to 29°C) and presumably other environmental conditions. It relates individual clearance rate ($CR_{ind}$; L $h^{-1}$) to dry weight (DW).

$$\text{CR}_{\text{ind}} = 6.79 \times \text{DW}^{0.73} \quad (4)$$

A comparison of the mean individual feeding rates predicted using Equation 4 for the same sizes of oysters on our

study reefs (converted from shell height to individual dry weight using Eq. 3) allowed a direct comparison of our field-measured data (Eq. 2) with previously determined laboratory-feeding rates.

A final equation was developed to provide a "simplest case scenario" model of how seston uptake is related to bivalve density ($N$; # $m^{-2}$), individual clearance rate ($CR_{ind}$; L $h^{-1}$ converted to $m^3$ $h^{-1}$), reef bottom area ($A_{reef}$; $m^2$), cross-section of water column over the reef ($A_{xsec}$; $m^2$), and water flow speed ($v$; cm $s^{-1}$ converted to m $h^{-1}$):

$$\text{Seston Uptake(\% Chl } a \text{ removal)} = (N \times CR_{ind} \times A_{reef})/(A_{xsec} \times v) \times 100 \quad (5)$$

Changes in seston concentration (i.e., seston removal) over short time periods caused by active suspension feeders in shallow, well-mixed water is essentially determined by feeding rate relative to water volume (Koseff et al. 1993; Haamer and Rodhe 2000; Riisgård et al. 2007). Equation 5 adds water flow to the relationship, thus providing a simple model that predicts the percent of the water column potentially cleared by an oyster reef in shallow, well-mixed tidal flows under optimum environmental conditions (e.g., temperature, dissolved oxygen concentration). The outputs from this model were compared to the actual measured chl *a* uptake (Eq. 1) for each of the eight study reefs using published individual clearance rates ($CR_{ind}$) from previous laboratory studies (Powell et al. 1992; Riisgård 1988).

## Results

The datasets from all eight reefs were collected under similar environmental conditions. Mean water depth ranged from 26 to 42 cm (all measurements were made in water depths <60 cm), mean flow speed from 2.2 to 13.5 cm $s^{-1}$, and flow length between 5 to 18 m (Table 2). For the five reefs with available water-quality data, temperature ranged from 22 to 27°C and salinity ranged from 24 to 32 psu for the time periods when seston data were collected. *C. virginica* was the dominant bivalve species on most of the study reefs, but hard clams (*M. mercenaria*) were abundant on one (the constructed reef in Two Sisters Creek, ACE Basin "Clam/oyster" reef). Ribbed mussels (*G. demissa*) occurred in low numbers on most reefs. Total bivalve densities (all species combined) ranged from 217 to 2831 $m^{-2}$ and mean shell heights ranged from 22.5 to 51.4 mm (Table 2).

The in situ fluorometry data showed total chl *a* removal ranging from −9.8% to 27.9% compared to −9.3% to 40.7% for the laboratory chl *a* data for the eight study reefs over time intervals ranging from 0.3 to 1.3 h (Fig. 1). The overall means for all study reefs combined were quite similar: 12.9% chl *a* removal for in situ fluorometry, and 13.6% for laboratory analysis of pumped water samples. However, there was wide variability between the two datasets for some reefs (e.g., Fig. 1: Woodland Creek, Oak Creek SC Oyster Restoration and Enhancement or SCORE reef constructed in 2001). A scatter plot of the means of in situ fluorometry data from the same time intervals (~5 min each) that water was pumped for the laboratory chl *a* measurements showed a strong linear relationship overall but wide variability in some cases (Fig. 2).

It should be noted that one of the datasets (Fig. 1; Store Creek, Natural Reef 1) showed a consistent increase in chl *a* concentration downstream compared to upstream over most of the measurement period. This was due to a persistent onshore wind that probably was re-suspending bottom sediments during the measurement period. One in situ fluorometry dataset was also collected over newly planted oyster shell with no live oysters within the reef (or other visible suspension feeders) to test for possible chl *a* removal by physical processes alone. This short-term (~10 min) set of measurements showed no differences in chl *a* concentrations between the downstream and upstream fluorometers.

Five of the study reefs were constructed using oyster shell, with subsequent development from natural spat set; one reef included seeded juvenile hard clams when constructed (Anderson and Yianopoulos 2003, Table 1). Two reef sites were in ACE Basin NERR (both constructed in 2000), two were in Oak Creek (one constructed in 2001, one in 2003), and another was located in Woodland Creek (constructed in 2003). The two oldest reefs had high removal rates, 15.1% and 23.0%. However, one of the youngest reefs had similarly high removal (21.8%). The two remaining reefs (constructed in 2001 and 2003) had removal rates <5% (Table 2; Fig. 1). These data cannot be used to infer a strong relation between constructed reef age and water-quality impacts because the reefs also varied in bivalve density and size, but they do show the expected overall trend of older reefs processing more water.

The overall mean *predicted* % cleared by the "simplest case scenario" model (Eq. 5) compared well to the *measured* % cleared only when the Powell et al. (1992) "low gear" individual feeding rates were used: 11.7% and 12.9%, respectively (Table 3). However, there was wide variability in some cases.

## Discussion

The most commonly used methods for measuring bivalve field feeding rates (seston removal) involve extrapolations from laboratory analyses of seston parameters (e.g., Peterson and Black 1991; Judge et al. 1992; Haamer and

**Table 2** Summary of environmental data for each study reef, measured chl *a* removal based on *in situ* fluorometry, calculated individual clearance rates, and predicted individual clearance rates based on equations in Riisgård (1988) and Powell et al. (1992)

| Date | Site | Mean bivalve density (# $m^{-2}$) | Mean bivalve shell size (mm) | Calculated[a] mean dry wt. (g) | Reef surface area ($m^2$) | Mean water cross-section over reef ($m^2$) | Mean water depth (cm) | Water flow length (m) | Mean water flow speed ($cms^{-1}$) | Mean water flow across reef ($m^3\ h^{-1}$) |
|---|---|---|---|---|---|---|---|---|---|---|
| 05/17/05 | Oak Ck SCORE #1 (2001) | 1500 | 44.0 | 0.11 | 15 | 1.08 | 36 | 5.0 | 13.5 | 525 |
| 05/17/05 | Oak Ck SCORE #2 (2003) | 1100 | 28.0 | 0.04 | 15 | 0.78 | 26 | 5.0 | 8.5 | 239 |
| 05/19/05 | ACE Clam/Oyster (2000) | 412 | 39.4 | 0.09 | 46 | 2.76 | 42 | 7.0 | 5.0 | 497 |
| 05/19/05 | ACE Recycled Shell (2000) | 2931 | 22.5 | 0.03 | 28 | 0.85 | 35 | 11.5 | 10.0 | 307 |
| 05/16/05 | Woodland Ck (2003) | 1662 | 39.6 | 0.09 | 75 | 1.95 | 39 | 15.0 | 4.7 | 330 |
| 06/07/06 | Store Ck#1-Natural | 217 | 31.8 | 0.06 | 72 | 1.28 | 32 | 18.0 | 2.2 | 101 |
| 06/07/06 | Store Ck#2-Natural | 922 | 31.7 | 0.05 | 72 | 1.56 | 39 | 18.0 | 8.5 | 477 |
| 06/08/06 | Bailey Ck-Natural | 820 | 51.4 | 0.16 | 75 | 1.90 | 38 | 15.0 | 2.8 | 192 |
| Means | | 1195.5 | 36.1 | 0.07 | 49.8 | 1.5 | 35.9 | 11.8 | 6.9 | 333.4 |

| Date | Site | Measured[b] min. chl *a* removal (% cleared) | Measured[b] max. chl *a* removal (% cleared) | Measured[b] mean chl *a* removal (% cleared) | Calculated[c] min. ind. Clearance ($l\ h^{-1}$) | Calculated[c] max. ind. Clearance ($l\ h^{-1}$) | Calculated[c] mean ind. Clearance ($l\ h^{-1}$) | Riisgård (1988) predicted ind. Clearance[d] ($l\ h^{-1}$) | Powell et al. (1992) clearance[e] "low gear" | Predicted ind. ($l\ h^{-1}$) "high gear" |
|---|---|---|---|---|---|---|---|---|---|---|
| 05/17/05 | Oak Ck SCORE #1 (2001) | 0.0 | 6.7 | 1.4 | 0.0 | 1.6 | 0.3 | 1.4 | 0.9 | 5.7 |
| 05/17/05 | Oak Ck SCORE #2 (2003) | 0.0 | 6.7 | 3.2 | 0.0 | 1.0 | 0.5 | 0.7 | 0.4 | 2.8 |
| 05/19/05 | ACE Clam/Oyster (2000) | 15.4 | 31.0 | 23.0 | 4.0 | 8.1 | 6.0 | 1.2 | 0.7 | 4.9 |
| 05/19/05 | ACE Recycled Shell (2000) | 9.5 | 20.0 | 15.1 | 0.4 | 0.7 | 0.6 | 0.5 | 0.3 | 1.9 |
| 05/16/05 | Woodland Ck (2003) | 8.3 | 28.5 | 21.8 | 0.2 | 0.8 | 0.6 | 1.2 | 0.7 | 4.9 |
| 06/07/06 | Store Ck#1-Natural | −15.7 | 4.8 | −9.8 | −1.0 | 0.3 | −0.6 | 0.8 | 0.5 | 3.5 |
| 06/07/06 | Store Ck#2-Natural | 16.1 | 22.2 | 20.2 | 1.2 | 1.6 | 1.5 | 0.8 | 0.5 | 3.5 |
| 06/08/06 | Bailey Ck-Natural | −18.7 | 35.7 | 27.9 | −0.6 | 1.1 | 0.9 | 1.8 | 1.1 | 7.2 |
| Means | | 1.9 | 19.5 | 12.9 | 0.52 | 1.90 | 1.21 | 1.00 | 0.65 | 4.30 |

[a] Soft tissue DW calculated using Coen (unpub. data): $DW=0.00003\times\text{shell height}^{2.1727}$

[b] Measured % cleared, see Eq. 1 in text; Min. and Max values based on % cleared calculated at 5-min intervals (see Fig. 2 for raw data)

[c] Calculated individual clearance, see Eq. 2 in text

[d] Riisgård (1988) clearance rates predicted using ind. $\text{clearance}=6.79\times DW^{0.73}$

[e] Powell et al. (1992) predicted individual clearance rates determined using "low gear" and "high gear" models for mean bivalve shell size on each reef.

Year of construction shown in parentheses for restored reefs.

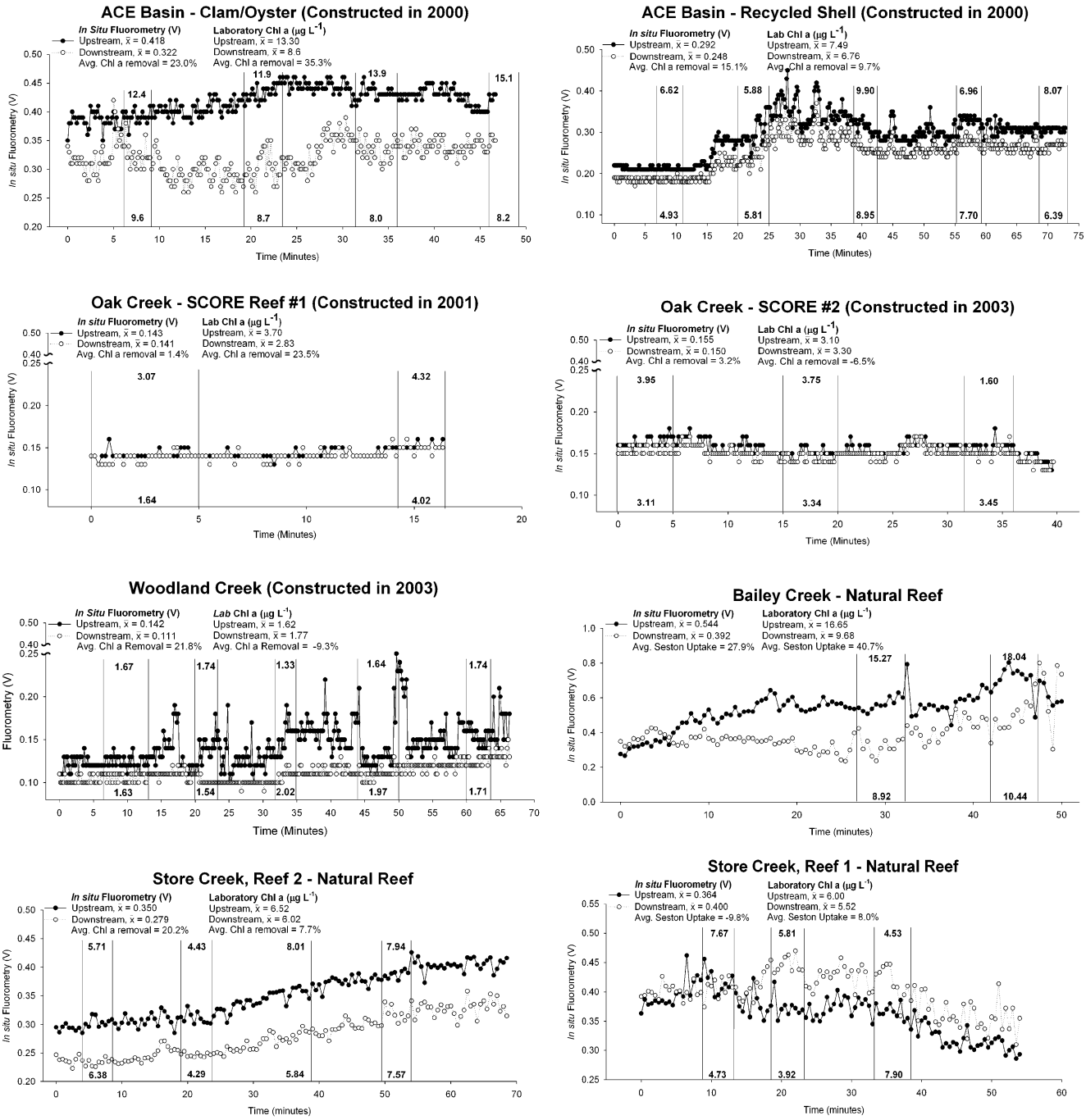



**Fig. 1** Plots of in situ fluorometry chl *a* data and laboratory chl *a* data from pumped water samples (shown as *numbers* between vertical lines) for all eight study reefs

Rodhe 2000; Cressman et al. 2003; Nelson et al. 2004). The approach used in the present study and more fully described in Grizzle et al. (2006) differs from previous efforts mainly by providing in situ measurements, along with laboratory analyses of discrete water samples. Our long-term aim is to develop in situ techniques that replace or at least can act as surrogates in the field complementing more costly and limited approaches that require taking discrete water samples for later laboratory analysis. The major advantage of an in situ approach sampling at one or more heights above the filter-feeding population is its ability to greatly increase spatial and temporal resolution of feeding effects because of the greatly increased sample sizes practical with in situ methods.

In order to provide comparative data, we measured chl *a* removal rates using both techniques with samples taken concurrently. Both techniques showed very similar mean chl *a* removal (12.9% vs. 13.6%), and a sample-by-sample

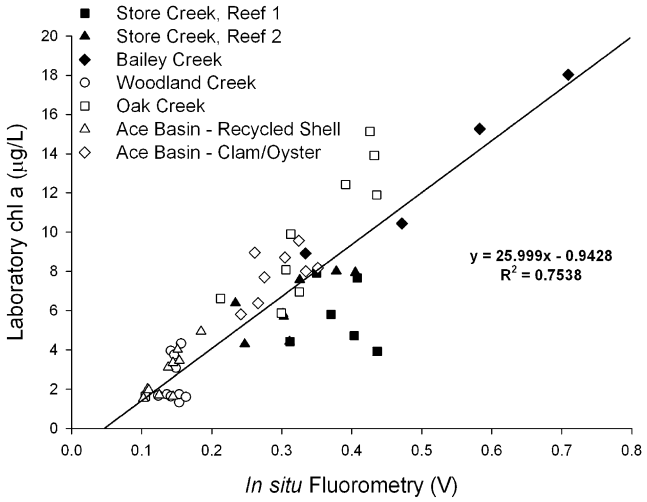



**Fig. 2** Relationship between in situ fluorometry data and laboratory chl *a* data (see Fig. 1 for raw data)

comparison of the combined 2005 and 2006 datasets showed a strong linear relationship (Fig. 2). Thus, overall, while both techniques on average yielded similar findings, we often observed wide variability between the in situ fluorometry and pumped water/laboratory chl *a* values for some individual reef datasets. This is not surprising since much of the variability can be attributed to differences in sample size (and frequency) and the integrative vs. discrete sampling differences between the two techniques, where a small amount of resuspended sediment with high microphytobenthos concentrations at these intertidal sites often can skew the results significantly (Coen et al. 2000). On an individual reef basis, sample size ranged from two to five pumped water samples analyzed in the laboratory compared to a range of 99 to 439 data points for the *in situ* fluorometry sampling (Fig. 1). This difference points to the major advantage of in situ methods compared to pumped water samples: temporal variability is "dampened out" due to the much larger number of measurements that can be made in situ.

Our study included natural and constructed reefs of varying known and unknown ages populated by different filter-feeding bivalve (primarily oyster) densities so that inferences could be drawn regarding the potential effects of constructed reefs on water quality, a major potential ecosystem service of shellfish-dominated reefs.

Total seston removal by a reef results from the feeding activities of all suspension feeders present and potentially the physical effects of flow variations across the reef (e.g., Dame et al. 1992; Porter et al. 2004a, b). Constructed reefs in South Carolina vary widely in size and density of their bivalve shellfish populations. Oysters typically dominate, but other taxa such as ribbed mussels (*G. demissa*) and hard clams (*M. mercenaria*) can be present in low to high numbers (Coen and Bolton-Warberg 2005; Coen et al. 2004, 2006; Walters and Coen 2006). Measurable seston removal would be expected soon after construction, and rates should increase over time as the density and size of suspension feeders on the reef increase. Measured seston removal rates on the constructed reefs were not clearly related to age of the reef (Table 2, Fig. 1), probably because bivalve size and density were not related just to age and because one of the reefs was seeded with juvenile hard clams during initial construction. Seston removal was, however, related to mean size and density of the bivalve populations, similar to the findings of Nelson et al. (2004). These data indicate that restored shellfish reefs should provide water-quality improvements soon after construction, and the overall impact is probably mainly determined by the size, density, and species composition of the resident filter-feeding populations relative to water flow characteristics over the reef (Coen et al. 2007, Newell et al. 2007).

## Our Field Data Compared to Previous Laboratory Studies

Individual feeding rates for bivalves typically have been measured in the laboratory with a few individuals and under carefully controlled conditions. A variety of methods have been used and there is substantial controversy regarding how the available published data should be interpreted (e.g., see Riisgård 2001 and responses by Cranford 2001 and Widdows 2001). Moreover, some factors that can strongly affect feeding behavior have not been simulated in the laboratory (Harsh and Luckenbach 1999). Thus, it has often been argued that field studies are needed to provide a more ecologically realistic understanding of bivalve feeding behavior (Powell et al. 1992; Dame and Libes 1993; Newell and Langdon 1996; Harsh and Luckenbach 1999; Judge et al. 1993; Cranford 2001; Porter et al. 2004b). Our primary objective for the present study was to measure chl *a* removal by oyster reefs which consisted mainly of *C. virginica*, but also included other bivalves and potentially other suspension feeders. Our quadrat sampling, however, indicated that *C. virginica* was the dominant suspension-feeder on all eight study reefs. Thus, the average individual bivalve clearance rates calculated for our study reefs (Eq. 2) mainly reflect oyster feeding

The mean individual clearance rates calculated from our in situ fluorometry seston removal measurements ranged from −0.6 to 6.0 L $h^{-1}$, with an overall mean of 1.21 L $h^{-1}$ (Table 2). Mean oyster size on our reefs ranged from 28.0 to 51.4 mm, with an overall mean of 36.1 mm (Table 2). Most laboratory-feeding rates for *C. virginica* have been determined using larger animals with the resulting clearance rates expressed relative to soft tissue dry weight (DW). Although these differences prohibit direct comparisons, approximations can be made. Recent modeling studies used clearance rates standardized to an oyster of 1 g dry weight

**Table 3** Comparison of measured whole-reef seston uptake (Eq. 1 in text) with predicted whole-reef seston uptake (% Chl a removal $=(N \times CR_{ind} \times A_{reef})/(A_{xsec} \times v) \times 100$; Eq. 5 in text) using individual clearance rates (variable B) predicted by the "low gear" and "high gear" models in Powell et al. (1992) and the model in Riisgård (1988)

| Date | Reef | Bivalve density ($N$; # $m^{-2}$) | Bivalve shell height (mm) | Predicted individual clearance rate[a] ($CR_{ind}$; L $ind^{-1}$ $h^{-1}$) | | | Reef bottom area ($A_{reef;}$ $m^2$) | Water cross-section ($A_{xsec}$; $m^2$) | Water flow speed ($v$; cm $s^{-1}$) | Water flow rate ($A_{xsec} \times v$; $m^3$ $h^{-1}$) | Measured[b] min. seston uptake (% Cleared) | Measured[b] max. seston uptake (% Cleared) | Measured[b] mean seston uptake (% Cleared) | Predicted[c] seston uptake (% cleared) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Powell et al. (1992) | | Riisgård (1988) | | | | | | | | Powell et al. (1992) | | Riisgård (1988) |
| | | | | "low gear" | "high gear" | | | | | | | | | "low gear" | "high gear" | |
| 05/17/05 | Oak Ck SCORE #1 (2001) | 1500 | 44.0 | 0.9 | 4.3 | 1.4 | 15 | 1.1 | 13.5 | 525 | 0.0 | 6.7 | 1.4 | 3.7 | 18.4 | 6.0 |
| 05/17/05 | Oak Ck SCORE #2 (2003) | 1100 | 28.0 | 0.4 | 2.3 | 0.7 | 15 | 0.8 | 8.5 | 239 | 0.0 | 6.7 | 3.2 | 3.0 | 15.6 | 4.8 |
| 05/19/05 | ACE Clam/Oyster (2000) | 412 | 39.4 | 0.7 | 3.7 | 1.2 | 46 | 2.8 | 5.0 | 497 | 15.4 | 31.0 | 23.0 | 2.8 | 14.2 | 4.6 |
| 05/19/05 | ACE Recycled Shell (2000) | 2931 | 22.5 | 0.3 | 1.6 | 0.5 | 28 | 0.9 | 10.0 | 306 | 9.5 | 20.0 | 15.1 | 8.5 | 42.2 | 13.4 |
| 05/16/05 | Woodland Ck (2003) | 1662 | 39.6 | 0.7 | 3.7 | 1.2 | 75 | 2.0 | 4.7 | 330 | 8.3 | 28.5 | 21.8 | 27.7 | 141.1 | 45.3 |
| 06/07/06 | Store Ck#1-Natural | 217 | 31.8 | 0.5 | 2.7 | 0.8 | 72 | 1.3 | 2.2 | 101 | −15.7 | 4.8 | −9.8 | 8.2 | 42.3 | 12.3 |
| 6/7/06 | Store Ck#2-Natural | 922 | 31.7 | 0.5 | 2.7 | 0.8 | 72 | 1.6 | 8.5 | 477 | 16.1 | 22.2 | 20.2 | 7.3 | 38.0 | 11.1 |
| 6/8/06 | Bailey Ck-Natural | 820 | 51.4 | 1.1 | 5.3 | 1.8 | 75 | 1.9 | 2.8 | 192 | −18.7 | 35.7 | 27.9 | 34.8 | 168.7 | 57.8 |
| Means | | 1196 | 36.1 | 0.7 | 3.3 | 1.1 | 49.8 | 1.5 | 6.9 | 333.3 | 1.9 | 19.5 | 12.9 | 12.0 | 60.1 | 19.4 |

[a] Individual clearance rates predicted using models in Powell et al. (1992) and Riisgård (1988) that relate clearance to bivalve size.
[b] Measured whole-reef seston uptake using Eq. 1 in text; minimum and maximum values based on % cleared calculated at 5-min intervals (see Fig. 2 for plots of raw data).
[c] Predicted whole-reef seston uptake from Eq. 1 in text for each of three difference individual clearance rates (see note 1).
Note that some units given in table must be converted to units used in Eq. 5: $CR_{ind}$ must be converted to $m^3$ $ind^{-1}$ $h^{-1}$; $v$ must be converted to m $h^{-1}$

(Newell and Koch 2004; Newell et al. 2005; Cerco and Noel 2007; Fulford et al. 2007), which corresponds to ~76 mm shell height (Newell et al. 2005). Because we did not measure dry weight of oysters on our study reefs when the seston uptake measurements were made, comparisons to previous laboratory studies require two standardizations: shell height to DW, then DW to clearance rate, as described above and involving Eqs. 2, 3, and 4.

In a recent comparative study that included data from multiple sites including subtidal and intertidal reefs in Virginia and collected over several years, Ross and Luckenbach (2006) developed a general power function ($DW = 0.00003 \times \text{shell height}^{2.3952}$; $R^2 = 0.80$) that yields 0.161 g DW for our average oyster with 36.1 mm shell height. Their model predicts that the "standard" 1 g DW oyster used in recent modeling studies would have a shell height of 77 mm (Newell and Koch 2004 indicated that an oyster with 1 g DW averages ~76 mm shell height). Ross and Luckenbach (2006) found substantial variability in best-fit models among their study sites, probably reflecting differences in environmental characteristics, oyster condition, and other factors. Therefore, it can be expected that oysters in different geographic areas will vary from the Ross and Luckenbach (2006) model. So, we used a power function developed using only South Carolina intertidal oysters (Eq. 3) that yields 0.07 g DW (less than half that predicted by the Ross and Luckenbach model) for our average 36.1 mm intertidal oyster. It is important to note that while the oysters on our study reefs were approximately half the shell height of the "standard" oyster (76 mm) used in most recent modeling studies, their DW probably was <10% of the "standard" (1 g DW) oyster.

To complete the standardization, we used the power function (Eq. 4) developed by Riisgård (1988) for oysters feeding under near optimal temperatures (27 to 29°C) and presumably other environmental conditions. This model predicts that our average oyster (36.1 mm; 0.07 g DW) would have a clearance rate of 1.00 L $h^{-1}$, compared to our calculated overall mean of 1.21 L $h^{-1}$. Although these values are quite close, it should be noted that there was wide variability among the means calculated on a reef-by-reef basis (Table 2). Also, the Riisgård (1988) power function relating DW to clearance has not been used in recent ecosystem modeling efforts. Newell and Koch (2004) measured clearance rates for subtidal oysters from 80 to 100 mm shell height and standardized their rates to 1 g DW. Their measurements at 25°C ranged from 7.05 to 9.58 L $h^{-1}$. Using the Ross and Luckenbach (2006) model, a 90-mm oyster would have a DW of 1.44 g. For an oyster of this weight, the Riisgård (1988) model predicts a clearance rate of 8.9 L $h^{-1}$, which is within the range measured by Newell and Koch (2004). Newell et al. (2005) standardized the data from Jordan (1987) to a 1-g DW oyster. For water temperatures from 25 to 27°C, they used clearance rates of 7.46 to 9.62 L $h^{-1}$. The Riisgård (1988) model predicts a clearance rate of 6.79 L $h^{-1}$ for the "standard" 1 g DW subtidal oyster, slightly lower than Jordan's (1987) data. Note several other recent modeling studies were based on the subtidal oyster data by Jordan (1987) standardized to different units (Cerco and Noel 2007; Fulford et al. 2007).

A final comparison to previous laboratory studies can be made using the extensive review by Powell et al. (1992). They found two general trends in the relation between bivalve size and feeding rate, which they termed "low gear" and "high gear." Based mainly on their own physiological energetics modeling simulations, they argued that the "low gear" values more likely represent 'typical' field feeding rates. Our calculated mean individual clearance rates compared quite well with the "low gear" rates of Powell et al. (1992), which on average are only about 15% of the high-gear rates for animals of a given shell size (Table 2). In a laboratory flume study, clearance rates for *C. virginica* were also observed to most closely match the Powell et al. "low gear" values (Harsh and Luckenbach 1999). The "low gear" rate predicted for a 1 g DW oyster is 2.0 L $h^{-1}$ compared to 12.7 L $h^{-1}$ for the "high gear" value. As discussed above, the rates used in recent modeling studies fall between 7.05 and 9.62 L $h^{-1}$, intermediate between the low- and high-gear predictions.

This analysis suggests two important conclusions. First, feeding rates determined by three different laboratory studies for the "standard" 1 g DW subtidal eastern oyster are quite similar: 9.6 L $h^{-1}$ (Jordan 1987); 8.9 L $h^{-1}$ (Riisgård 1988); and 8.4 L $h^{-1}$ (Newell and Koch 2004). Thus, the relationship between oyster DW and clearance rates modeled by the power function of Riisgård (1988) seems reasonably valid for allowing a comparison of our data for small intertidal oysters (<0.2 g DW) to recent modeling studies based on the "standard" 1 g DW oyster. Secondly, the wide temporal variability in seston removal measured over our study reefs suggests that intertidal oyster feeding rates in nature may be *on average* substantially lower than the *maximum rates* reported from some laboratory studies, as suggested by others (e.g., Powell et al. 1992; Harsh and Luckenbach 1999; Cranford 2001; also see more discussion below).

## Future Studies and the Value of our Approach

The most pressing need overall is to quantify seston removal by oyster reefs over as wide a range of environmental conditions as practical so that ecologically realistic seston removal rates can be determined on appropriate temporal scales (monthly, seasonally, etc.). The field flume ("tunnel") studies by Dame and colleagues

are examples of the kind of research needed, but using more robust in situ methods that do not isolate the study populations from components of their environment (e.g., wind waves and complex non-directional flows; see Judge et al. 1993) that potentially affect feeding. Future studies should focus on factors that are most important and need to be considered in ecosystem-level models (Newell et al. 2005, 2007; Pomeroy et al. 2006; Cerco and Noel 2007; Fulford et al. 2007).

Sufficient data exist to warrant more careful consideration of tidal currents and wind-generated waves. For example, Dame et al. (1985) reported increased seston concentrations downstream of an intertidal oyster reef when tidal current speeds exceeded 15 cm $s^{-1}$, indicating a net export of seston from the reef. On an annual basis, ebb tide uptake of chl *a* measured over 33 tidal cycles was only ~43% of the flood tide rate; Dame et al. (1992) hypothesized that lower flood tide current speeds that resulted in less sediment resuspension was partly the cause. Tidal flow speeds in the present study were only measured at one height (5 to 10 cm above the bottom), so no data are available on details of tidal flow characteristics. These fixed-height measurements (which on average ranged from 5.0 to 13.5 cm $s^{-1}$), however, reflected flow speed at the surface of each reef. Unfortunately, existing studies on the effects of water flow speed on eastern oyster feeding and growth have yielded conflicting data and are not sufficient to characterize the range of effects (Grizzle et al. 1992; Lenihan et al. 1996; Harsh and Luckenbach 1999; Porter et al. 2004a, b). One reason these studies have been inconclusive may be a secondary effect of currents on sediment re-suspension or re-distribution near the bottom, which would make direct inferences on feeding effects difficult (Harsh and Luckenbach 1999). The exact causal mechanisms for how water flow affects populations of oysters remain to be determined, but it is well established that water flow at some speed becomes inhibitory to individual bivalve feeding and growth (Wildish and Kristmanson 1997; Peterson 2001; Grizzle et al. 2001).

Our data indicate that wind-generated waves that disturb the bottom can cause an increase in chl *a* concentration as water flows across a reef, similar to the findings by Dame et al. (1985, 1992) for tidal currents, and as discussed above. On two occasions (Fig. 1; Store Creek, Natural Reef 1; and early in the measurement period for Bailey Creek) wind waves caused substantial re-suspension of bottom sediments along the flow path and resulted in increased chl *a* concentrations downstream, causing a net export of chlorophyll from the reef. Although these data do not necessarily indicate inhibited feeding, they do suggest a net export of seston by oyster reefs under some conditions. We are aware of no previous research on how waves affect oyster feeding, but they can cause re-suspension and increased transport of deposited sediments from reefs in areas exposed to wind and/or boat-generated waves (Grizzle et al. 2002; Wall et al. 2005). Such events may also result in removal of sediments accumulating on oyster reefs at much higher rates than occur under more typical (and better studied) non-storm conditions. Separating the effects of bivalve feeding responses from physical factors directly affecting sediment erosion and transport will be difficult, probably requiring some combination of laboratory and field studies.

In addition to hydrodynamic factors, others have suggested behavioral responses that might result in decreased oyster feeding rates in the field compared to laboratory studies. Harsh and Luckenbach (1999) suggested that the wide temporal variations in feeding activity typically observed among individual oysters in a population should be considered in order to determine more accurate reef-level (population average) impacts. Powell et al. (1992) argued based on physiological energetic considerations that their "low gear" model probably represented the most ecologically realistic feeding rate for most bivalves. Additional factors that could cause decreased feeding rates in the field include parasites and predators. When considering the complexity of the oyster's natural environment, including both physical and biological factors, it seems reasonable to expect lower average feeding rates than typically have been measured in the laboratory.

Our "simplest case scenario" model (Eq. 5) was developed initially to predict the potential effects of a small constructed oyster reef on water quality. It is mainly offered as a tool for estimating the potential short-term impact of a population of suspension-feeding bivalves in shallow, well-mixed waters. As noted above, it includes the minimum number of variables for relating seston depletion to bivalve feeding (Haamer and Rodhe 2000; Riisgård et al. 2007). Thus, if the model's major assumptions are met—a well-mixed water column and minimal horizontal transport of seston into or out of the modeled area—it should accurately predict seston removal. The fact that the model accurately predicted mean % seston (chl *a*) cleared for all eight study reefs combined (12.9% measured uptake vs. 12.0% predicted; Table 3) indicates this is the case, at least when the Powell et al. (1992) "low gear" individual feeding rate is assumed. Although the overall means were quite similar, however, only three of the eight individual reef predictions closely matched the measured seston uptake. In one case (ACE clam/oyster reef), the measured uptake was nearly double the predicted rate even using the Powell et al. "high gear" assumed individual feeding rate. Measured minimum and maximum seston uptake were widely variable, in four cases including no uptake or negative values. These data strongly suggest widely variable feeding rates over short time periods, but other factors may be involved.

These observations suggest two important conclusions relative to the utility of our model. First, the model in its present form (Eq. 5) is capable of accurately predicting seston uptake—*if* its major assumptions are met and appropriate individual clearance ($CR_{ind}$) rates are used. Therefore, one immediate application might be for estimating the amount of water constructed shellfish reefs in shallow, well-mixed tidal flows would potentially filter. Secondly, departures from these conditions will require models with increasing complexity (see Grizzle et al. 2001; Porter et al. 2004a, b; Riisgård et al. 2007). Recent ecosystem models involving the eastern oyster have included equations relating major environmental variables such as salinity and temperature to oyster feeding rates (Cerco and Noel 2007; Fulford et al. 2007). Adding these functions to our model would broaden its application potential.

In conclusion, it has been repeatedly argued that there is a need to measure oyster feeding rates under ecologically realistic conditions in the field to compare with laboratory measurements. This has become increasingly important as ecosystem-level models involving oyster feeding have proliferated, in large measure because of the emphasis by managers and others on the potential role of oysters in water-quality improvements. The present study and other recent field studies confirm the substantial filtration capabilities of oyster reefs. And inferred individual clearance rates for oysters on our study reefs compared well to some previous laboratory data. However, the wide temporal variability in seston removal measured over our study reefs (and reported previously by others for laboratory-feeding studies) suggests that oyster feeding rates in nature may be *on average* substantially lower than the *maximum rates* reported from some laboratory studies. More research on the feeding behavior of oysters in nature including a wide range of size classes and under various environmental conditions is badly needed.

**Acknowledgments** Funding and support was provided by South Carolina Sea Grant (#NA16RG2250, #NA03NMF4630272), NOAA's Community Restoration Program, the South Carolina Department of Natural Resources, and the ACE Basin NERR. We thank Nancy Hadley, Amy Dukes, Charlie Zemp, Michael Hodges, Andy Hollis and Amanda Powers for assistance with logistics, field site selection, coordination, and field and/or laboratory analyses. This is Contribution #469 from the University of New Hampshire, Jackson Estuarine Laboratory Contribution Series and #640 from the Marine Resources Research Institute, South Carolina Department of Natural Resources.

## References

Ackerman, J.D., M.R. Loewen, and P.F. Hamblin. 2001. Benthic–Pelagic coupling over a zebra mussel reef in western Lake Erie. *Limnology and Oceanography* 46: 892–904.

Alpine, A.E., and J.E. Cloern. 1992. Trophic interactions and direct physical effects control phytoplankton biomass and production in an estuary. *Limnology and Oceanography* 37: 946–955.

Anderson, W.D., and G.M. Yianopoulos. 2003. Final Report to NOAA-RC, Intertidal oyster and clam bed restoration in the ACE Basin, South Carolina using recycled oyster shell, hatchery raised clam seed, and other cultch material, Project Period, July 2000–September 2002, 18pp.

Arar, E.J. 1997. Method 447.0, Determination of chlorophylls a and b and identification of other pigments of interest in marine and freshwater algae using high performance liquid chromatography with visible wavelength detection. National Exposure Research Laboratory, Office of Research and Development, U.S. Environmental Protection Agency, Cincinnati, Ohio. 20 pp.

Brumbaugh, R.D., L.A. Sorabella, C.O. Garcia, W.J. Goldsborough, and J.A. Wesson. 2000. Making a case for community-based oyster restoration: an example from Hampton Roads, Virginia, U. S.A. *Journal of Shellfish Research* 19: 467–472.

Cerco, C.F., and M.R. Noel. 2007. Can oyster restoration reverse cultural euthrophicaiton in Chesapeake Bay? *Estuaries and Coasts* 30: 331–343.

Coen, L.D., and M. Bolton-Warberg. 2005. An examination of the impacts of various harvesting/ management strategies and boat wakes on oyster bed condition, recovery rates, shoreline erosion and restoration. A Marine Recreational Fisheries Stamp Program Final Report, 96 pp.

Coen, L.D., and M.W. Luckenbach. 2000. Developing success criteria and goals for evaluating oyster reef restoration: ecological function or resource exploitation? *Ecological Engineering* 15: 323–343. doi:10.1016/S0925-8574(00)00084-7.

Coen, L.D., M. Judge, C. Moncreiff, and K. Hammerstrom. 2000. Hard clam (*Mercenaria mercenaria*) mariculture in U.S. waters: evaluating the effects of large-scale field growout practices on clam growth, nutrition and inshore estuarine creek communities. Final Project Report to Saltonstall-Kennedy, NMFS. 38 pp.

Coen, L.D., M. Bolton-Warberg, Y. Bobo, D. Richardson, A.H. Ringwood, and G.I. Scott. 2004. Oyster Beds. In *A baseline assessment of environmental and biological conditions in the May River, Beaufort County, South Carolina: final report submitted to the Town of Bluffton*, eds. R.F. Van Dolah, D.M. Sanger, and A.B. Filipowicz, eds. 127–147.

Coen, L.D., M. Bolton-Warberg, and J.A. Stephen. 2006. An examination of oyster reefs as biologically-critical estuarine ecosystems. Final Report, Grant R/ER-10, Submitted to the South Carolina Sea Grant Consortium, 214pp. plus appendices.

Coen, L.D., R.D. Brumbaugh, D. Bushek, R. Grizzle, M.W. Luckenbach, M.H. Posey, S.P. Powers, and G. Tolley. 2007. AS WE SEE IT. A broader view of ecosystem services related to oyster restoration. *Marine Ecology. Progress Series* 341: 303–307. doi:10.3354/meps341299.

Coughlan, J. 1969. The estimation of filtering rate from the clearance of suspensions. *Marine Biology* 2: 356–368. doi:10.1007/BF00355716.

Cranford, P.J. 2001. Evaluating the 'reliability' of filtration rate measurements in bivalves. *Marine Ecology Progress Series* 215: 303–305. doi:10.3354/meps215303.

Cressman, K.A., M.H. Posey, M.A. Mallin, L.A. Leonard, and T.D. Alphin. 2003. Effects of oyster reefs on water quality in a tidal creek estuary. *Journal of Shellfish Research* 22: 753–762.

Dame, R. 1996. *Ecology of marine bivalves: an ecosystem approach*. 254. Boca Raton, FL: CRC.

Dame, R., and S. Libes. 1993. Oyster reefs and nutrient retention in tidal creeks. *Journal of Experimental Marine Biology and Ecology* 171: 251–258. doi:10.1016/0022-0981(93)90007-B.

Dame, R.F., T.G. Wolaver, and S.M. Libes. 1985. The summer uptake and release of nitrogen by an intertidal oyster reef. *Netherlands*

*Journal of Sea Research* 19: 265–268. doi:10.1016/0077-7579(85)90032-8.

Dame, R., N. Dankers, T. Prins, H. Jongma, and A. Smaal. 1991. The influence of mussel beds on nutrients in the Western Wadden Sea and Eastern Scheldt Estuaries. *Estuaries* 14: 130–138. doi:10.2307/1351686.

Dame, R.F., J.D. Spurrier, and R.G. Zingmark. 1992. In situ metabolism of an oyster reef. *Journal of Experimental Marine Biology and Ecology* 164: 147–159. doi:10.1016/0022-0981(92)90171-6.

Dame, R., D. Bushek, and T. Prins. 2001. The role of suspension feeders as ecosystem transformers in shallow coastal environments. In *The ecology of sedimentary coasts*, ed. K. Reise, 11–37. Berlin: Springer.

French McCay, D.P., C.H. Peterson, J.T. DeAlteris, and J. Catena. 2003. Restoration that targets function as opposed to structure: replacing lost bivalve production and filtration. *Marine Ecology Progress Series* 264: 197–212. doi:10.3354/meps264197.

Fulford, R.S., D.L. Breitburg, R.I.E. Newell, W.M. Kemp, and M. Luckenbach. 2007. Effects of oyster population restoration strategies on phytoplankton biomass in Chesapeake Bay: a flexible modeling approach. *Marine Ecology Progress Series* 336: 43–61. doi:10.3354/meps336043.

Grabowski, J.H., and C.H. Peterson. 2007. Restoring oyster reefs to recover ecosystem services. In *Ecosystem engineers: concepts, theory and applications*, eds. K. Cuddington, J.E. Byers, W.G. Wilson, and A. Hastings, 281–298. Amsterdam: Elsevier–Academic.

Grant, J., G. Bugden, E. Horne, M.-C. Archambault, and M. Carreau. 2007. Remote sensing of particle depletion by coastal suspension-feeders. *Canadian Journal of Fisheries and Aquatic Science* 64: 387–390. doi:10.1139/F07-021.

Grizzle, R.E., R. Langan, and W.H. Howell. 1992. Growth responses of suspension-feeding bivalve molluscs to changes in water flow: differences between siphonate and nonsiphonate taxa. *Journal of Experimental Marine Biology and Ecology* 162: 213–228. doi:10.1016/0022-0981(92)90202-L.

Grizzle, R.E., M.V. Bricelj, and S.E. Shumway. 2001. Physiological ecology of *Mercenaria mercenaria.* pp. 305–382 In *Biology of the hard clam*, eds. J.N. Kraeuter and M. Castagna. Elsevier.

Grizzle, R.E., J.R. Adams, and L.J. Walters. 2002. Historical changes in intertidal oyster (*Crassostrea virginica*) reefs in a Florida lagoon potentially related to boating activities. *Journal of Shellfish Research* 21: 749–756.

Grizzle, R.E., J.K. Greene, M.W. Luckenbach, and L.D. Coen. 2006. A new *in situ* method for measuring seston uptake by suspension-feeding bivalve mollusks. *Journal of Shellfish Research* 25: 643–649. doi:10.2983/0730-8000(2006)25[643:ANISMF]2.0.CO;2.

Haamer, J. 1996. Improving water quality in a eutrophied fiord system with mussel farming. *Ambio* 25: 356–362.

Haamer, J., and J. Rodhe. 2000. Mussel *Mytilus edulis* (L.) filtering of the Baltic Sea outflow through the Öresund—an example of a natural, large-scale ecosystem restoration. *Journal of Shellfish Research* 19: 413–421.

Harsh, D.A., and M.W. Luckenbach. 1999. Materials processing by oysters in patches: interactive roles of current speed and seston composition. In *Oyster reef habitat restoration: a synopsis and synthesis of approaches*, eds. M.W. Luckenbach, R. Mann, and J.A. Wesson, 251–265. Gloucester Point, Virginia: Virginia Institute of Marine Science.

Jordan, S.J. 1987. Sedimentation and remineralization associated with biodeposition by the American oyster *Crassostrea virginica* (Gmelin). Doctoral dissertation. University of Maryland. College Park.

Judge, M.L., L.D. Coen, and K.L. Heck Jr. 1992. The effect of long-term alteration of in situ water currents on the growth of the hard clam, *Mercenaria mercenaria*, in the northern Gulf of Mexico. *Limnology and Oceanography* 37: 1550–1559.

Judge, M.L., L.D. Coen, and K.L. Heck Jr. 1993. Does *Mercenaria mercenaria* encounter elevated food levels in seagrass beds? Results from a novel technique to collect suspended food resources. *Marine Ecology Progress Series* 92: 141–150. doi:10.3354/meps092141.

Kennedy, V.S. 1996. The ecological role of the eastern oyster, *Crassostrea virginica*, with remarks on disease. *Journal of Shellfish Research* 15: 177–183.

Koseff, J.R., J.K. Holen, S.G. Monismith, and J.E. Cloern. 1993. Coupled effects of vertical mixing and benthic grazing on phytoplankton populations in shallow, turbid estuaries. *Journal of Marine Research* 51: 843–868. doi:10.1357/0022240933223954.

Lenihan, H.S., C.H. Peterson, and J.M. Allen. 1996. Does flow speed also have a direct effect on growth of active suspension-feeders: an experimental test on oysters. *Limnology Oceanography* 41: 1359–1366.

Luckenbach, M.W., R. Mann, and J.A. Wesson, eds. 1999. Oyster Reef Habitat Restoration. A Synopsis and Synthesis of Approaches. Virginia Institute of Marine Science. Virginia Institute of Marine Science Press, Gloucester Point, VA, 358 pp.

Luckenbach, M.W., L.D. Coen, P.G. Ross Jr., and J.A. Stephen. 2005. Oyster reef habitat restoration: relationships between oyster abundance and community development based on two studies in Virginia and South Carolina. *Journal of Coastal Research* (Special Issue) 40: 64–78.

Navarro, J.M., and L.A. Velasco. 2003. Comparison of two methods for measuring filtration rate in filter feeding bivalves. *Journal of the Marine Biological Association of the United Kingdom* 83: 553–558. doi:10.1017/S0025315403007471h.

Nelson, K.A., L.A. Leonard, M.H. Posey, T.D. Alphin, and M.A. Mallin. 2004. Using transplanted oyster (*Crassostrea virginica*) beds to improve water quality in small tidal creeks: a pilot study. *Journal of Experimental Marine Biology and Ecology* 298: 347–368. doi:10.1016/S0022-0981(03)00367-8.

Newell, R.I.E. 1988. Ecological changes in Chesapeake Bay: Are they the result of overharvesting the Eastern oyster (*Crassostrea virginica*)? In *Understanding the estuary: advances in Chesapeake Bay research*, ed. M. Lynch, 536–546. Gloucester Point, Virginia: Chesapeake Research Consortium Publication 129.

Newell, R.I.E., and E.W. Koch. 2004. Modeling Seagrass density and distribution in response to changes in turbidity stemming from bivalve filtration and seagrass sediment stabilization. *Estuaries* 27: 793–806.

Newell, R.I.E., and C.J. Langdon. 1996. Mechanisms and physiology of larval and adult feeding. In *The Eastern Oyster Crassostrea virginica*, eds. V.S. Kennedy, R.I.E. Newell, and A.F. Eble, 185–229. College Park, Maryland: Maryland Sea Grant College, University of Maryland.

Newell, R.I.E., T.R. Fisher, R.R. Holyoke, and J.C. Cornwell. 2004. Influence of eastern oysters on nitrogen and phosphorus regeneration in Chesapeake Bay, USA. In *The comparative roles of suspension feeders in ecosystems. NATO Science Series: IV—Earth and Environmental Sciences*, eds. R. Dame, and S. Olenin, 93–120. Netherlands: Springer.

Newell, R.I.E., T.R. Fisher, R.R. Holyoke, and J.C. Cornwell. 2005. Influence of eastern oysters on N and P regeneration in Chesapeake Bay, USA. In *The comparative roles of suspension feeders in ecosystems. NATO science series: IV—earth and environmental sciences*, eds. R. Dame, and S. Olenin, 93–120. Netherlands: Kluwer.

Newell, R.I.E., W.M. Kemp, J.D. Hagy, C.F. Cerco, J.M. Testa, and W.R. Boynton. 2007. Top-down control of phytoplankton by oysters in Chesapeake Bay, USA: Comment on Pomeroy et al. (2006). *Marine Ecology Progress Series* 341: 293–298. doi:10.3354/meps341293.

Peterson, C.H. 2001. Integrating nutritional physiology and ecology to explain interactions between physics and biology in *Mercenaria*

*mercenaria*. In *Biology of the hard clam*, eds. J.N. Kraeuter, and M. Castagna, 423–439. Elsevier.

Peterson, C.H., and R. Black. 1991. Preliminary evidence for progressive sestonic food depletion in incoming tide over a broad tidal sand flat. *Estuarine, Coastal and Shelf Science* 32: 405–413. doi:10.1016/0272-7714(91)90052-D.

Pomeroy, L.R., C.F. D'Elia, and L.C. Schaffner. 2006. Limits to top-down control of phytoplankton by oysters in Chesapeake Bay. *Marine Ecology Progress Series* 325: 301–309. doi:10.3354/meps325301.

Porter, E.T., J.C. Cornwell, and L.P. Sanford. 2004a. Effect of oysters *Crassostrea virginica* and bottom shear velocity on benthic–pelagic coupling and estuarine water quality. *Marine Ecology Progress Series* 27: 61–75. doi:10.3354/meps271061.

Porter, E.T., L.P. Sanford, G. Gust, and F.S. Porter. 2004b. Combined water-column mixing and benthic boundary-layer flow in mesocosms: key for realistic benthic–pelagic coupling studies. *Marine Ecology Progress Series* 271: 43–60. doi:10.3354/meps271043.

Powell, E.N., E.E. Hofmann, J.M. Klinck, and S.M. Ray. 1992. Modeling oyster populations. I. A commentary on filtration rate. Is faster always better? *Journal of Shellfish Research* 11: 387–398.

Riisgård, H.U. 1988. Efficiency of particle retention and filtration rate in 6 species of Northeast American bivalves. *Marine Ecology Progress Series* 45: 217–223. doi:10.3354/meps045217.

Riisgård, H.U. 2001. On measurement of filtration rates in bivalves—the stony road to reliable data: review and interpretation. *Marine Ecology Progress Series* 211: 275–291. doi:10.3354/meps211275.

Riisgård, H.U., J. Lassen, M. Kortegaard, L.F. Møller, M. Friedrichs, M.H. Jensen, and P.S. Larsen. 2007. Interplay between filter-feeding zoobenthos and hydrodynamics in the shallow Odense Fjord (Denmark)—earlier and recent studies, perspectives and modeling. *Estuarine, Coastal and Shelf Science* 75: 281–295. doi:10.1016/j.ecss.2007.04.032.

Ross, P.G., and M.W. Luckenbach. 2006. Relationships between shell height and dry tissue biomass for the eastern oyster (*Crassostrea virginica*). 9th International Conference on Shellfish Restoration, Charleston, SC, November 2006.

Shumway, S. 1996. Environmental factors. In *The Eastern Oyster Crassostrea virginica*, eds. V.S. Kennedy, R.I.E. Newell, and A.F. Eble, 467–514. College Park: Maryland Sea Grant College, University of Maryland System.

Wall, L.M., L.J. Walters, R.E. Grizzle, and P.E. Sacks. 2005. Recreational boating activity and its impact on the recruitment and survival of the oyster *Crassostrea virginica* on intertidal reefs in Mosquito Lagoon, Florida. *Journal of Shellfish Research* 24: 965–973.

Walters, K., and L.D. Coen. 2006. A comparison of statistical approaches to analyzing community convergence between natural and constructed oyster reefs. *Journal of Experimental Marine Biology and Ecology* 330: 81–95. doi:10.1016/j.jembe.2005.12.018.

Widdows, J. 2001. Bivalve clearance rates: inaccurate measurements or inaccurate reviews and misrepresentations? *Marine Ecology Progress Series* 221: 303–305. doi:10.3354/meps221303.

Wildish, D.J., and D.D. Kristmanson. 1997. Benthic suspension feeders and flow. Cambridge University Press.

Wilson-Ormond, E.A., E.N. Powell, and S.M. Ray. 1997. Short-term and small-scale variation in food availability to natural oyster populations: food, flow and flux. *Marine Ecology* 18: 1–34. doi:10.1111/j.1439-0485.1997.tb00424.x.